Stream Identification and **Stream Reservation** are two of the most commonly confused ideas in **Time-Sensitive Networking (TSN)**.

They are related, but they answer two different questions:

- **Stream Identification** asks: **"Which stream does this arriving frame belong to?"**
- **Stream Reservation** asks: **"Has the network committed bandwidth, buffering, and forwarding resources for that stream end-to-end?"**

Those are not the same operation. A bridge can identify a stream without having reserved it, and a stream can be reserved on paper but still be mishandled if arriving packets are not identified correctly.

First, what is a "stream" in TSN?

A **stream** is not just "some packets on Ethernet." It is a set of frames that the network treats as one traffic object for quality of service, filtering, scheduling, shaping, and sometimes redundancy handling. In IEEE 802.1 terminology, a bridge or end station commonly maps packets to an internal label called a **stream_handle**; IEEE 802.1CB uses that stream_handle for replication and elimination, and IEEE 802.1Q uses it for quality-of-service functions. Importantly, IEEE 802.1 does **not** use the stream_handle itself to choose the output port; forwarding/path selection is a different function.

That distinction is foundational:

- **Forwarding** decides **where** the frame goes.
- **Stream identification** decides **what stream it belongs to**.
- **Reservation** decides **what resources that stream is allowed to consume**.

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1. Stream Identification

1.1 Why stream identification is needed

Suppose a switch receives a frame. The switch cannot apply per-stream policing, gating, shaping, or reliability logic unless it first determines which stream that frame belongs to. VLAN priority alone is not enough, because many different applications may share the same priority or VLAN. IEEE 802.1CBdb's project description explicitly states that stream identification is required by Layer 2 traffic-management mechanisms such as ingress policing, traffic scheduling, congestion management, and mapping to traffic classes. (1.ieee802.org)

So the switch performs a **classification step**:

Look at selected packet fields and/or the ingress port, then assign an internal stream_handle.

Once that happens, later TSN functions can operate on that stream_handle. IEEE material notes that a frame can then be sent to a meter, a time gate, a packet-length filter, a timed byte-count filter, or be mapped to a class-of-service queue.

1.2 What fields can be used?

IEEE's TSN stream-identification overview lists the kinds of fields that can determine the stream_handle. These include:

- ingress port number
- source Media Access Control (MAC) address
- destination MAC address
- VLAN identifier
- Layer 2 priority
- Internet Protocol (IP) source address
- IP destination address
- next protocol such as User Datagram Protocol (UDP) or Transmission Control Protocol (TCP)
- source port number
- destination port number
- Differentiated Services Code Point (DSCP).

That means stream identification can be as simple as "frames with destination MAC X and VLAN Y" or as specific as "UDP packets from source A to destination B on port 1722 with DSCP Z arriving on ingress port 3."

1.3 The key internal object: `stream_handle`

Think of the **stream_handle** as the switch's internal name for a stream.

A frame comes in with some headers. The stream-identification function matches it and says:

"Inside this bridge, this packet is stream 27."

From that point onward, the bridge can apply stream-specific treatment. IEEE 802.1CB uses the stream_handle to:

- add a Layer 2 sequence number,
- replicate packets,
- eliminate duplicates based on sequence numbers,
- and rewrite destination MAC and VLAN information if needed. IEEE 802.1Q uses it for gates, filters, meters, and queue assignment.

So, in practical terms:

[
\text{frame headers} ;\longrightarrow; \text{stream_handle} ;\longrightarrow; \text{TSN treatment}
]

1.4 Passive and active stream identification

There are two very different styles.

Passive stream identification

This means the bridge **recognizes** the stream but does not change the frame headers as part of identification. It is just classification. IEEE draft text for extended stream identification shows passive forms such as:

- Null Stream identification
- Source MAC and VLAN Stream identification
- IP Stream identification.

Active stream identification

This means the bridge identifies the stream and can also **rewrite selected header fields** so that downstream functions can handle it in a particular way. The IEEE material shows an **Active Destination MAC and VLAN Stream identification** function that examines destination address and VLAN identifier and can overwrite destination address, VLAN identifier, and priority.

This is subtle but important. Passive identification says “this is stream A.” Active identification says “this is stream A, and I will now stamp the frame with the destination MAC, VLAN, and priority convention that downstream TSN processing expects.”

1.5 Standard identification types

IEEE material around 802.1CB and its extension work shows these families of identification functions:

- **Null Stream identification**: based on destination address and VLAN identifier.
- **Source MAC and VLAN Stream identification**: based on source address and VLAN identifier.
- **Active Destination MAC and VLAN Stream identification**: active mapping that can overwrite destination address, VLAN, and priority.
- **IP Stream identification**: uses Ethernet plus IP-layer fields.
- **Extended Stream identification**: mask-and-match over more arbitrary fields.

Extended stream identification exists because the original methods are not always rich enough for industrial, automotive, or other application-specific traffic patterns. IEEE's 802.1CBdb project page says the existing methods in 802.1CB were insufficient for some applications and that the amendment adds new identification functions. (1.ieee802.org)

1.6 What happens after a stream is identified?

Once a frame has a stream_handle, the network can do several things with it.

In IEEE 802.1Q

The stream_handle can send the frame to:

- a flow meter,
- a time-scheduled gate,
- a packet-length filter,
- a timed total-byte-count filter,
- or an internal priority value that maps to a queue.

In IEEE 802.1CB

The stream_handle can:

- add a Layer 2 sequence number,
- replicate the frame into multiple copies,
- pass it through packet-elimination state machines,
- remove the sequence tag,
- rewrite destination MAC and VLAN values.

So stream identification is the **front door** into the TSN/FRER pipeline.

1.7 Example 1: a simple camera stream

Imagine a zonal camera sends video frames to a central compute unit.

Assume the frames arrive on:

- ingress port 5,
- VLAN 100,
- destination MAC `01-00-5E-...`,
- priority 5.

The bridge may use `{ingress port, destination MAC, VLAN}` to assign `stream_handle = 12`. From then on, all those frames are treated as one stream: they can be metered, mapped to the correct queue, shaped, and filtered as a group. Another camera on the same VLAN but

different source MAC or ingress port can map to a different stream_handle even though both use the same priority. That is exactly why stream identification exists.

1.8 Example 2: unicast application traffic that needs header normalization

Suppose an end station emits several kinds of application traffic, all as ordinary unicast Ethernet. Downstream switches want a clean convention where one application stream is always seen as a particular `{destination MAC, VLAN, priority}` triplet. IEEE's extended stream-identification draft explicitly gives the example that extended stream identification can be coupled with active destination MAC and VLAN stream identification to assign a specific `{MAC address, VLAN, priority}` triplet to packets belonging to a particular unicast application stream identified by selected fields in the MAC service data unit.

In plain language:

- first, identify the application stream using deeper packet fields,
- then rewrite the outer Ethernet marking so the rest of the network can treat it consistently.

1.9 Example 3: seamless redundancy with frame replication and elimination

Suppose a safety-critical control stream must be sent redundantly over two disjoint paths. IEEE 802.1CB specifies identification and replication of frames for redundant transmission, identification of duplicates, and elimination of duplicate frames. The stream_handle can send the frame through a sequence-generation function, add a Layer 2 sequence number, cause replication, and later enable elimination of duplicates based on sequence numbers. (1.ieee802.org)

So the sequence is:

1. identify frame as stream S,
2. assign stream_handle S,
3. replicate it to path A and path B,
4. tag each copy with sequence information,
5. downstream node recognizes both as the same stream and discards the duplicate.

Without stream identification, the network would not know which frames belong to the same redundant stream.

1.10 A very important misconception

One important misconception is that if a frame is identified as stream X, that tells the switch which port to forward it to. IEEE's TSN stream-identification overview explicitly says there is no provision in IEEE 802.1 for using the stream_handle to select on which port(s) to output a packet.

So do not confuse:

- **classification** with
- **forwarding/path control**.

A stream_handle is for **treatment**, not for basic Layer 2 destination lookup.

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2. Stream Reservation

2.1 What problem reservation solves

Even if the bridge can identify a stream perfectly, that does not mean the network has enough resources to carry it with bounded latency and low loss. IEEE 802.1Qat says network resources along the entire data path must be assured **before transmission takes place**, and that this requires traffic-stream descriptors plus a signaling protocol to carry the reservation information along the end-to-end path.

That is the essence of stream reservation:

Before the talker starts sending a time-sensitive stream, the network checks and commits the required resources hop by hop.

Resources here include, at minimum:

- egress bandwidth,
- queueing/shaping capacity,
- buffering and admission state,
- sometimes path and class selection.

2.2 Classic distributed reservation: SRP / MSRP

The original distributed reservation mechanism is **SRP** (Stream Reservation Protocol), built around **MSRP** (Multiple Stream Reservation Protocol). IEEE 802.1Qat describes SRP as a signaling protocol for the end-to-end management of resource reservation for guaranteed-quality streams, including registration, deregistration, and retention of reservation information.

At a high level the actors are:

- **Talker**: source of the stream
- **Listener**: receiver of the stream
- **Bridges**: intermediate nodes that evaluate and maintain reservation state

The normal case described in IEEE material is:

1. the **Talker** makes an offering declaration in the talker-to-listener direction,
2. the **Listener** responds with a Ready declaration,
3. queues are configured as registration progresses from listener back toward the talker,
4. acknowledgement from listener to talker must return before data can be transmitted.

That is very different from best-effort Ethernet, where you just start sending.

2.3 What the Talker advertises

IEEE's stream-reservation example shows the Talker Advertise carrying, among other things:

- stream ID,
- accumulated latency,
- frame length,
- interval.

As the declaration propagates through bridges, the accumulated latency is incremented by each bridge's latency contribution. That gives the listener and the network a running picture of the end-to-end latency budget.

So the advertisement is not just "I want to talk." It is more like:

"I am stream 0x1234. I will send frames of this size at this interval. My current path latency budget has accumulated to this amount."

2.4 What the Listener does

If the listener is ready to receive the stream, it sends a **Listener Ready** declaration. IEEE material also shows that, on receipt of a Talker Advertise, the listener first makes an MVRP VLAN membership request for the VLAN named in the talker advertisement and then sends MSRP Listener Ready for the stream.

That order matters. The listener is not only saying "yes, I want the stream," but also ensuring the VLAN context is present so the stream can actually be carried.

2.5 What each bridge does

This is where the real reservation logic lives.

The bridge performs bandwidth analysis per transmit port. IEEE material states that bandwidth reservation is done on the **TX side**, defines an **Active Listener** as Listener Ready or Listener Ready Failed, and shows that the bridge includes per-stream information such as:

- Stream ID,
- Talker Attribute (Advertise or Failed),
- Talker Traffic Class (A or B),
- TSpec,
- Stream destination MAC,
- Stream VID,
- and an Approved/Disapproved flag set by Qav.

So the bridge is doing something like this for each egress port:

1. Which streams might exit this port?
2. Which of them actually have active listeners downstream?
3. What bandwidth and timing do they require?
4. If I admit this new stream, do I still satisfy the already admitted ones?
5. If yes, approve and propagate the reservation state; if not, mark failure.

This is **admission control** at Layer 2.

2.6 Why shaping is tied to reservation

Reservation without queue control is meaningless. IEEE 802.1Qav provides forwarding and queueing enhancements for time-sensitive streams and says it uses timing derived from IEEE 802.1AS. It enables end-to-end quality-of-service guarantees negotiated over SRP to be realized in a bridged LAN.

So SRP says, “reserve this stream,” and Qav gives the bridge the queueing/shaping behavior needed to actually honor that reservation. They are different pieces of the same machine.

2.7 The classic AVB budget protection idea

In the classic Audio Video Bridging (AVB) model, IEEE tutorial material explains that only **75%** of the link bandwidth can be reserved for Stream Reservation classes. The reason given is to protect both best-effort traffic and the reserved streams themselves, because allowing more than that would break the latency guarantee.

So if a 100 Mbit/s link is carrying SR-class traffic in that classic model, you do not reserve 99 Mbit/s for streams. You stop earlier, because determinism requires margin.

That is a good intuition pump: **reservation is not “use every last bit”; it is “use only what can still be guaranteed.”**

2.8 Example 4: distributed reservation for a camera stream

Let us make this concrete.

A camera wants to send one frame of 1500 bytes every 250 microseconds.

Very roughly:

$$
1500 \times 8 = 12000 \text{ bits}
$$

every (250 ,\mu s), which is about:

$$
\frac{12000}{250 \times 10^{-6}} = 48 \text{ Mbit/s}
$$

before detailed Ethernet overheads.

Now the talker sends a Talker Advertise with:

- stream ID,
- frame length,
- interval,
- initial accumulated latency.

Each bridge on the path:

- checks the egress port that would carry the stream,
- asks whether another ~48 Mbit/s in the chosen traffic class is still admissible,
- updates accumulated latency,
- and only if the port remains within the stream-reservation budget does it propagate approval.

When the listener sends Listener Ready back, the queue configuration and bandwidth reservation converge from listener toward talker. Only after that should time-sensitive transmission begin. That is the logic IEEE documents describe.

2.9 What happens if there is not enough capacity?

Then the stream is not admitted cleanly. The standard ecosystem includes failed/asking-failed states; IEEE material lists TalkerFailed, ListenerReadyFailed, and ListenerAskingFailed among the events that can trigger bandwidth reallocation.

Conceptually, the network is saying:

"I know what stream this is, but I cannot guarantee its requested service on this path in the current state."

This is a huge difference from normal Ethernet, which would simply accept the traffic and let congestion happen later.

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3. 802.1Qcc: enhanced and centralized reservation models

The first-generation distributed SRP model works, but it is not always the best fit for large TSN systems. IEEE 802.1Qcc extends SRP to support more streams, configurable stream-reservation classes, better stream descriptions, Layer 3 streaming, deterministic convergence, and a **User/Network Interface (UNI)** for routing and reservations. (1.ieee802.org)

IEEE's TSN overview shows that 802.1Qcc defines three configuration models:

1. **Fully centralized**
2. **Fully distributed**
3. **Centralized network / distributed user**

3.1 Fully centralized model

In this model there is central intelligence, typically:

- **CUC**: Central User Configuration
- **CNC**: Central Network Configuration.

IEEE's overview says that in the fully centralized model, simple devices do not need to implement signaling protocols; requirements are exchanged over the UNI and the network returns success or failure.

This is very attractive in controlled automotive or industrial systems because the end devices can stay simple.

Example 5: central vehicle network planner

Imagine a vehicle with:

- four cameras,
- two radars,
- one lidar,
- one diagnostics logger.

A CNC knows the full topology and can compute:

- which path each stream takes,
- which VLAN and traffic class it uses,
- whether resources are sufficient,
- and potentially even transmission gate schemes or synchronization-related configuration.

IEEE's TSN overview explicitly notes that a CNC may configure resource-allocation classes, traffic classes, and potentially transmission gate schemes, VLANs, active topologies, synchronization, and TSN domains.

So instead of every endpoint shouting "I need bandwidth," the network planner computes the answer once and installs it.

3.2 Fully distributed model

In the fully distributed model there is no central entity. End stations and bridges must implement distributed protocols such as SRP and related registration mechanisms. IEEE's overview calls these more complex devices.

This is closer to the classic AVB/SRP style:

- talker advertises,
- listener joins,
- bridges propagate declarations and reserve resources.

3.3 Centralized network / distributed user model

This is a hybrid. IEEE's overview says the **network** has central intelligence through a CNC, but the **user side** still uses signaling at the UNI, such as SRP or RAP.

That means:

- endpoints still express their needs dynamically,
- but the network computes and installs the actual resource allocation centrally.

This is often a good engineering compromise.

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4. Stream Identification and Stream Reservation are different stages

This is the most important conceptual summary.

4.1 Stream identification is per-frame, at runtime

Every frame arrives and is classified into a stream_handle using ingress metadata and selected header fields. This is a **packet-processing** operation.

4.2 Stream reservation is per-stream, at setup/change/removal time

When an application wants to create a TSN stream, the network decides whether it can admit that stream and configures state along the path. This is a **resource-management** operation.

So one is “classify this frame now,” while the other is “admit this stream into the network and configure all bridges.”

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5. Their interaction

They only work properly when combined.

If you reserve but do not identify correctly

Suppose the network has admitted a camera stream, but arriving packets are misclassified because the bridge is matching the wrong VLAN or destination MAC. Then the reserved stream may fall into the wrong queue, miss its meter, or bypass the intended filters. The reservation exists, but the data plane is not using it.

If you identify but do not reserve

Suppose the bridge can identify the stream perfectly, but no admission control has been performed. Then the stream may still contend with other traffic and lose its latency guarantee.

In other words:

[
\text{Identification} + \text{Reservation} + \text{Queue/Shaping} = \text{Deterministic

treatment}
]

Not one of those alone.

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6. A useful mental picture

Think of an airport.

- **Stream Identification** is the check-in desk deciding which flight you belong to.
- **Stream Reservation** is whether a seat has actually been booked for that flight.
- **Shaping / scheduling / gates** are the boarding process and runway slots.

If you are identified but not reserved, you know your flight but may not have a seat.
If you are reserved but not identified correctly, the system may not send you to the right gate.

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7. Automotive reality check

This part matters for you.

AUTOSAR's current explanatory TSN document says that AUTOSAR does **not** support:

- time-aware shapers,
- active stream identification,
- frame replication,

and assumes VLAN-aware Ethernet switches. It also describes TSN as the IEEE 802 family that provides bounded low latency and high availability for automotive and industrial systems.

So in real automotive AUTOSAR deployments today, you often use only a **subset** of the full IEEE TSN toolbox. That means when you read IEEE 802.1CB/802.1Q as pure standards, you are reading a bigger capability set than what many AUTOSAR systems actually instantiate.

That is especially important for your topic here:

- IEEE supports richer stream identification machinery than many automotive stacks presently expose.
- IEEE supports richer reservation/configuration models than many in-vehicle software stacks currently automate.

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8. Crisp distinction to keep forever

Here is the clean version.

Stream Identification

Maps an arriving packet to an internal stream object.

Typical question:

“Using ingress port, MAC/IP fields, VLAN, ports, or masks, which stream_handle is this?”

Typical outputs:

- stream_handle
- per-stream filters/meters/gates/queues
- optional FRER handling.

Stream Reservation

Allocates network resources for a stream end-to-end before or during service establishment.

Typical question:

“Can I guarantee the required bandwidth and latency for this stream on this path?”

Typical outputs:

- admitted or rejected
- bridge state installed
- queue/shaper reservation
- stream class/path/topology configuration.

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9. Final summary

A **TSN stream** is a traffic object that the network wants to treat coherently. **Stream Identification** is the per-frame classification process that converts header/port information into an internal stream_handle so the bridge can meter, gate, shape, filter, replicate, or recover the traffic. **Stream Reservation** is the per-stream control-plane process that determines whether the path has enough resources and then installs the corresponding state using SRP/MSRP or centralized 802.1Qcc-style configuration. One tells the bridge **what the frame is**; the other tells the network **what it has promised that stream**.